

JMeter Instructions:

JMeter Demo Steps (FastAPI Direct Endpoint)

1. Start the FastAPI application
Command:
`python3.11 -m uvicorn app.main:app --reload`
2. Open Apache JMeter
3. In the left panel, right-click the Test Plan
Add → Threads (Users) → Thread Group
4. Configure the Thread Group with:
 - Number of Threads (users): 100
 - Ramp-up period (seconds): 10
 - Loop Count: 1
5. Right-click the Thread Group
Add → Sampler → HTTP Request
6. Configure the HTTP Request:
 - Name: Create User Direct
 - Server Name or IP: 127.0.0.1
 - Port Number: 8000
 - Method: POST
 - Path: /users-direct
7. In the HTTP Request, use Body Data and paste:

```
{
  "operation": "CREATE_USER",
  "name": "Alice",
  "email": "alice@test.com",
  "age": 25
}
```
8. Right-click the HTTP Request
Add → Config Element → HTTP Header Manager
9. In HTTP Header Manager, add:
 - Name: Content-Type
 - Value: application/json
10. Right-click the Thread Group and add listeners:

- Add → Listener → View Results Tree
- Add → Listener → Summary Report
- Add → Listener → Aggregate Report
- Add → Listener → Response Time Graph

11. Click the green Start button in JMeter to run the test

12. Observe the results:

- View Results Tree: shows each request and response
- Summary Report: shows average response time, min, max, error %, throughput
- Aggregate Report: shows detailed performance statistics
- Response Time Graph: shows response time trend over time

13. Key metrics to note:

- # Samples = total number of requests
- Average = average response time in milliseconds
- Min = fastest request
- Max = slowest request
- Error % = percentage of failed requests
- Throughput = requests handled per second